Applied Surface Science 458 (2018) 438–445

Contents lists available at ScienceDirect

Applied Surface Science

journal homepage: www.elsevier.com/locate/apsusc

Full Length Article

Synthesis of peptide-containing calcium phosphate nanoparticles exhibiting highly selective adsorption of various proteins

Suzuka Kojima[a,b], Fukue Nagata[b], Shinichi Kugimiya[a], Katsuya Kato[b,*]

[a] Materials Chemistry Course, Graduate School of Engineering, Aichi Institute of Technology, 1247 Yachigusa, Yakusa-cho, Toyota 470-0392, Japan
[b] National Institute of Advance Industrial Science and Technology, 2266-78, Anagahora, Moriyamaku, Nagoya, Aichi 463-8560, Japan

ARTICLE INFO

Keywords:
Nanoparticles
Hydroxyapatite
Peptide
Protein
Selective adsorption
Recycling

ABSTRACT

Nanomaterials have small size and unique properties and are used as adsorbents in various fields. For biomaterials, it is necessary to use nanomaterials prepared from nontoxic molecules. Particularly, protein adsorption on hydroxyapatite, $Ca_{10}(PO_4)_6(OH)_2$ (HAp), which is highly biocompatible and ecological, was thoroughly studied by varying the reaction conditions and controlling the crystal faces. We easily prepared two peptide-containing HAp (peptide-HAp) materials with high selectivity for protein adsorption. The morphologies and characteristics of the peptide-HAp composites were similar to those of raw HAp; however, the surface charge was significantly changed by increasing the amount of peptides. The amount of cytochrome *c* adsorbed on peptide-HAp containing poly(L-glutamic acid) and of bovine serum albumin adsorbed on poly(L-lysine)-containing HAp significantly increased compared to that on pure HAp. In addition, as the amount of peptides was increased, peptide-HAp showed improved selectivity for protein adsorption by electrostatic interaction between peptide-HAp and protein. Moreover, peptide-HAp could be reused at least five times in adsorption-desorption cycles, and it was expected that the materials could be used as adsorbents and in separation columns.

1. Introduction

Recently, nanotechnologies have achieved growth in various fields, and depending on the preparation technique, nanoparticles, nanotubes, and nanofibers can be obtained [1–4]. These materials are approximately 1–100 nm in length, and their unique properties are applied in medical, commercial, and environmental fields. Specifically, nanoparticles have been used as absorbents and column materials for high-performance liquid chromatography (HPLC). Nanoparticles have unique structural, chemical, electrical, and mechanical properties; therefore, they provide ideal conditions for protein immobilization [5,6]. As general absorbents, ceramic nanomaterials such as silica are used widely in industrial fields; these absorbents are also improved by techniques such as surface modification [7,8]. Hikosaka et al. reported that DNA was adsorbed on mesoporous silica (MPS) with different pore sizes, which was functionalized with aminosilane, and the amount of DNA adsorbed on MPSs with diethylenetriamine increased with the pore size [9]. However, MPS materials are prepared by a complex process involving organic templates and surface functionalization; therefore, we focused on biocompatible and safe materials such as hydroxyapatite.

Hydroxyapatite, $Ca_{10}(PO_4)_6(OH)_2$, designated as HAp, is a calcium phosphate compound and a principal inorganic component of tooth and bone; therefore, it is highly biocompatible and ecological. HAp crystals are hexagonal and have two crystal faces. In addition, HAp is known to have strong affinity with proteins; thus, it is used as an adsorbent and column carrier material [10–12]. HAp can also adsorb various proteins with relatively low selectivity. Several researchers have reported the preparation of HAp with adsorption selectivity for proteins by varying the reaction conditions and controlling the crystal faces. However, it is difficult to obtain perfect selectivity by controlling only the particle structures of HAp [13,14]. Recently, studies were conducted on HAp-organic molecule composites, and the protein adsorption behavior on these materials was investigated. Xu et al. described the adsorption behavior of human serum albumin (HSA) and ovalbumin (OVA) on a hydroxyapatite/chitosan (HAp/CS) composite. They reported that a much higher amount of HSA could be adsorbed on the HAp/CS materials compared to OVA [15]. Thus, these HAp composites are potentially suitable as chromatographic tools.

Biopolymers, such as polypeptides, have many types of side chain

Abbreviations: HAp, hydroxyapatite; Cyt c, cytochrome *c*; MGB, myoglobin; BSA, bovine serum albumin; ANS, 8-anilino-1-naphthalenesulfonic acid; SDS-PAGE, sodium dodecyl sulfate-polyacrylamide gel electrophoresis

* Corresponding author.
E-mail addresses: suzuka-kojima@aist.go.jp (S. Kojima), katsuya-kato@aist.go.jp (K. Kato).

https://doi.org/10.1016/j.apsusc.2018.07.114
Received 2 April 2018; Received in revised form 10 July 2018; Accepted 17 July 2018
Available online 18 July 2018
0169-4332/ © 2018 Elsevier B.V. All rights reserved.

structures; therefore, they have different surface potentials and structures. Peptides also have good biocompatibility and a flexible molecular backbone [16–19]. We considered that combining peptides with HAp could change particle characteristics such as the surface potential of peptide-HAp composites.

The aim of this study was to easily synthesize new peptide-containing HAp particles exhibiting highly selective adsorption of various proteins. We used two types of peptides: poly(L-glutamic acid) with a carboxyl group and poly(L-lysine) with an amino group in their side chains. For protein adsorption experiments on peptide-HAp composites, bovine serum albumin as an acidic protein, myoglobin as a neutral protein, and cytochrome *c* as a basic protein were used to confirm the selectivity for protein adsorption. We focus on the effect of peptide combination with HAp on the selective adsorption of proteins. Additionally, the properties of synthesized HAp particles with respect to their use as adsorbents and carrier materials for column chromatography were evaluated.

2. Experimental

2.1. Materials

Diammonium hydrogen phosphate [$(NH_4)_2HPO_4$], calcium acetate monohydrate [$(CH_3COO)_2Ca{\cdot}H_2O$], ethanol, sodium chloride (NaCl), hydrogen peroxide (H_2O_2), 4-aminoantipyrine (4-AAP), and phenol crystals were obtained from Wako Pure Chemical Industries (Osaka, Japan). Bovine serum albumin [BSA; isoelectric point (pI) = 4.7, molecular weight (Mw) = 67,000 Da], cytochrome *c* from equine heart (Cyt c; pI = 10.0, Mw = 12,300 Da), myoglobin from equine skeletal muscle (MGB; pI = 7.0, Mw = 17,800 Da), and 8-anilino-1-naphthalenesulfonic acid (ANS) were purchased from Sigma-Aldrich Co. (St. Louis., MO). Two peptides [poly(L-glutamic acid sodium salt) (p-Glu; pI = approximately 2.4, Mw > 12,000) and poly(L-lysine hydrobromide) (p-Lys; pI = approximately 12.0, Mw > 12,000)] were obtained from Peptide Institute, Inc. (Osaka, Japan). EzApply, EzStain Aqua, e-PAGEL, and EzStandard Prestain Blue were purchased from ATTO Co. (Tokyo, Japan). The Bio-Rad Protein Assay was supplied by Bio-Rad Laboratories (Hercules, CA). All chemicals were of analytical grade and were used without further purification.

2.2. Synthesis of HAp nanoparticles

2.2.1. Hap preparation

HAp was synthesized according to the procedure described in our previous report [20]. First, 200 mL of 15 mM $(CH_3COO)_2Ca$ solution was added to 200 mL of 9 mM $(NH_4)_2HPO_4$ solution, and the mixture was stirred for 5 min at 20 °C. This mixture was heated to 60 °C (heating rate 1 °C/min) and stirred for a further 3 h at the same temperature [21]. The obtained products were collected by centrifugation at 6,000 rpm for 10 min. The solid material was recovered by filtration, washed with deionized water, and freeze-dried.

2.2.2. Peptide-HAp preparation

We prepared six types of peptide-HAp with different amounts of peptides, and each peptide-HAp was named according to the type and amount of peptides. In the case of p-Glu-containing HAp (p-Glu-HAp), p-Glu (5, 10, or 20 mg) was added to 200 mL of 15 mM $(CH_3COO)_2Ca$ solution. After stirring for 30 min at 20 °C, 200 mL of 9 mM $(NH_4)_2HPO_4$ solution was added to the mixture. This mixture was stirred for 5 min, and the following procedure was performed as described in Section 2.2.1. The procedure for preparing p-Lys-containing HAp (p-Lys-HAp) was as follows: first, p-Lys (5, 10, or 20 mg) was dissolved in 200 mL of 9 mM $(NH_4)_2HPO_4$ solution. Then, 200 mL of 15 mM $(CH_3COO)_2Ca$ solution was added to the mixture. The resulting mixture was treated using the same procedure as that for p-Glu-HAp.

2.3. Characterization of synthesized HAp

The morphologies of synthesized HAp were observed by field-emission scanning electron microscopy (FE-SEM; S-4300 instrument operated at 10.0 kV, Hitachi Ltd., Tokyo, Japan) and transmission electron microscopy (TEM; JEM-2010, JEOL Ltd., Tokyo, Japan) at an accelerating voltage of 200 kV. Scanning TEM (STEM) images were taken using a JEOL JEM-2100 Plus instrument (JEOL Ltd., Tokyo, Japan) operated at 200 kV. Fourier-transform infrared (FT-IR; FT/IR-4700, JASCO Co., Tokyo, Japan) spectroscopy was performed by the attenuated total reflection method. To evaluate the peptide content, thermogravimetry and differential thermal analysis (TG-DTA; Thermo Plus TG 8120, Rigaku Co., Tokyo, Japan) were performed. The heating temperature ranged from room temperature to 1,000 °C (heating rate 10 °C/min). The crystallinity of HAp was measured by powder X-ray diffraction (XRD; RINT2000/PC, Rigaku Co., Tokyo, Japan) using a CuKα source. XRD patterns were recorded over the 2θ range 3.0–60.0° (scan speed = 2.0°/min); the generator settings were 40 kV and 30 mA. The specific surface area and pore volume were determined by the Brunauer-Emmett-Teller (BET; TriStar 3000, Shimadzu Co., Kyoto, Japan) method and by obtaining nitrogen (N2) adsorption/desorption isotherms, respectively. The pore size distribution was calculated by the Barrett-Joyner-Halenda (BJH) method. To evaluate the molar ratio of calcium to phosphate (Ca/P), the materials were measured by inductively coupled plasma optical emission spectrometry (ICP-OES; IRIS Advantage, Thermo Fisher Scientific Inc., MA). The surface potential of HAp was determined with a ζ-potential analyzer (ELSZ-1000, Otsuka Electronics Co., Tokyo, Japan). Before the measurements, the particles were dispersed in 10 mM phosphate buffer (pH 7.0) with sonication for 3 min.

2.4. Protein adsorption on HAp

Protein adsorption was investigated using Cyt c, MGB, and BSA. Proteins were dissolved in 10 mM phosphate buffer (pH 7.0). The synthesized HAp (5 mg) was added to 1 mL of 500 μg/mL protein solution, and the mixture was stirred overnight at 20 °C. The supernatant was separated by centrifugation at 14,000 rpm for 5 min. The amount of protein adsorbed was determined by the Bradford protein assay with UV–vis spectroscopy (λ = 595 nm) (Infinite F200 PRO, Tecan Group Ltd., Männedorf, Switzerland).

2.5. Mechanisms of protein adsorption on HAp

To investigate protein adsorption behavior on peptide-HAp, the adsorption kinetics were calculated by fitting the equilibrium data to Langmuir and Freundlich models [22].

The Langmuir isotherm model assumes monolayer adsorption on the surface of peptide-HAp and is described by Eq. (1):

$$C_s/Q_e = C_s/Q_m + 1/(K_L Q_m), \tag{1}$$

where C_s is the equilibrium concentration (mg/mL), Q_e is the amount of protein adsorbed on peptide-HAp (mg/mg), Q_m is the maximum amount adsorbed (mg/mg), and K_L is the equilibrium constant (mL/mg).

The Freundlich isotherm model is suitable for heterogeneous systems and is represented by Eq. (2):

$$Q_e = K_F C_s^{1/n}, \tag{2}$$

where K_F is the equilibrium constant of this model (mg/mg) and $1/n$ is the Freundlich adsorption intensity constant.

2.6. Oxidation activity of Cyt c protein on HAp

The oxidation activity of free Cyt c and adsorbed Cyt c on HAp was investigated. Each HAp (5 mg) was added to a mixture of 100 μL Cyt c

solution (1 mg/mL) and 900 μL of 10 mM phosphate buffer (pH 7.0), and the mixture was stirred as described in Section 2.4. After centrifugation at 14,000 rpm for 5 min, 10 mM phosphate buffer (860 μL), 1.76% 4-AAP (20 μL), 6% phenol (20 μL), and 19.6 mM H_2O_2 (100 μL) were added and incubated at 30 °C for 10 min. The supernatant was separated by centrifugation at 14,000 rpm for 3 min, and the solution was measured by UV–vis spectrophotometry (DU 800, Beckman Coulter Inc., IN) at 500 nm.

2.7. Analysis of structural changes in the proteins adsorbed on peptide-HAp

Conformational changes in adsorbed protein on peptide-HAp were measured. Each peptide-HAp (1 mg) was mixed with a protein solution [Cyt c or BSA, 100 μg/100 μL in 10 mM phosphate buffer (pH 7.0)], and the mixture was stirred overnight. Subsequently, ANS solution (50 μg/500 μL) was added to the mixture. After stirring in the dark for 3 h at 20 °C, the fluorescence emission spectra of ANS-bound protein were recorded at excitation and fluorescence wavelengths of 387 nm and 400 to 600 nm, respectively, using a spectrofluorophotometer (RF-5300PC, Shimadzu Co., Kyoto, Japan).

2.8. Protein separation from protein mixture by peptide-HAp and cycle test

To adsorb one protein on peptide-HAp from a protein mixture and recover the adsorbed protein by desorption, the following experiment was performed. Each peptide-HAp (5 mg) was added to 1 mL of a protein mixture of Cyt c and BSA (500 μg of each protein), and the mixture was stirred for 1 h at 20 °C. The solid material was separated by centrifugation at 14,000 rpm for 5 min and washed once with 10 mM phosphate buffer (pH 7.0). After centrifugation, 1 mL of 1 M NaCl solution was added to the product. To evaluate the adsorbed protein, the suspension was stirred for 1 h, and the supernatant was separated by centrifugation. The proteins in the supernatant were evaluated using a sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) unit WSE-1100 (ATTO Co., Tokyo, Japan), with 10% polyacrylamide gel. The gel was stained with EzStain Aqua (ATTO Co., Tokyo, Japan).

When collection materials are used as recyclable adsorbents, cycling efficiency is an important factor. The experiment was performed using two peptide-HAp. The adsorption and desorption cycles were repeated five times. Other procedures were performed as described above.

3. Results and discussion

3.1. Morphology of synthesized HAp

The as-synthesized HAp was characterized using FE-SEM, TEM, BET surface area, and BJH pore size distribution. Fig. 1(A–C) and (a–c) shows FE-SEM and TEM images of the morphologies of HAp, p-Glu-HAp (10 mg), and p-Lys-HAp (10 mg) particles, respectively. All the synthesized samples showed rod-shaped and plate-like nanoparticles with particle sizes of approximately 60 × 30 nm. From these results, the samples of composites containing peptide showed a morphology that was visually similar to that of raw HAp. In addition, to evaluate the effect of the amount of peptides, we prepared p-Glu-HAp (Fig. S1) and p-Lys-HAp (Fig. S2) containing different amounts of peptides (5 and 20 mg). According to the results, the morphologies of these samples were preserved; thus, it was confirmed that the morphologies of the as-prepared HAp were not significantly influenced by the amount of peptides.

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.apsusc.2018.07.114.

Nitrogen (N2) adsorption-desorption isotherms of the synthesized p-Glu-HAp and p-Lys-HAp are shown in Fig. S3 and reveal type IV isotherms [23]. The specific surface area and pore volume of raw HAp were 96 m^2/g and 0.79 cm^3/g, respectively (Table 1). The specific surface areas of p-Glu-HAp with 5, 10, and 20 mg peptide were 107, 119, and 134 m^2/g, while the pore volumes were 0.91, 0.73, and 0.79 cm^3/g, respectively. For p-Lys-HAp (5, 10, and 20 mg), the surface areas were 106, 133, and 123 m^2/g, and the pore volumes were 0.92, 0.84, and 0.93 cm^3/g, respectively. For all the samples, the obtained values revealed that the structural characteristics of the synthesized HAp containing various amounts of peptides resembled those of raw HAp.

3.2. Characterization of peptide-HAp

To confirm the presence of peptides in nanoparticles, the elemental distribution of the synthesized peptide-HAp was investigated by STEM-EDX analysis. STEM-EDX images of p-Glu-HAp (20 mg) and p-Lys-HAp (20 mg) are shown in Fig. 2(a) and (b), respectively. The calcium, phosphorous, and nitrogen mapping images confirmed the presence of HAp (calcium and phosphorous) and the peptide (nitrogen) in the nanoparticles, as shown in Fig. 2. In addition, these mapping images

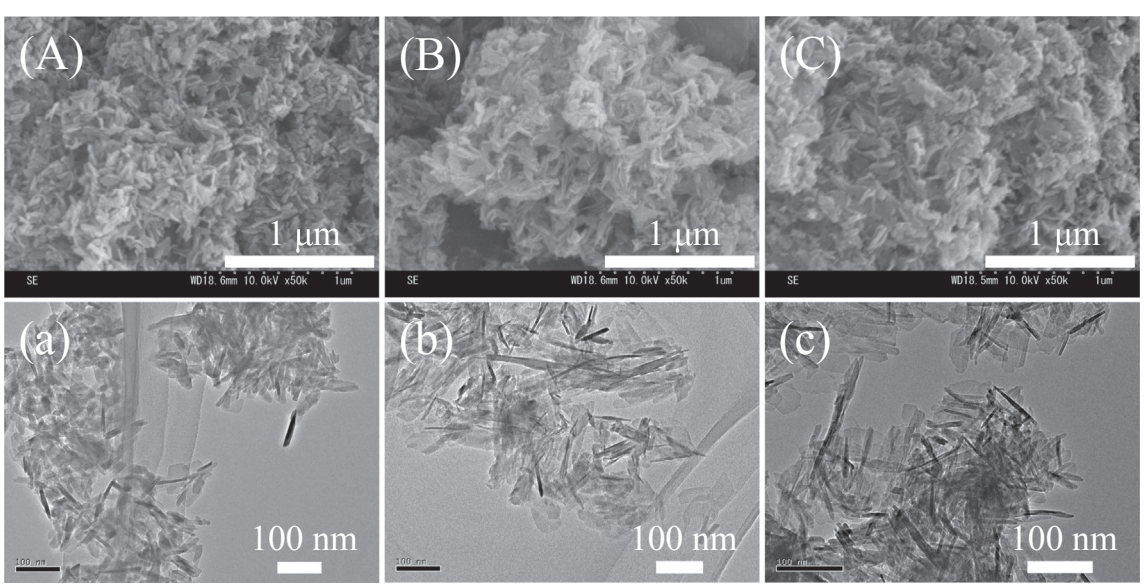

Fig. 1. FE-SEM (A–C) and TEM (a–c) images of synthesized HAp particles: (A, a) HAp, (B, b) p-Glu-HAp (10 mg), and (C, c) p-Lys-HAp (10 mg).

Table 1
Characterization of the synthesized HAp particles.

Sample	Surface area[(a)] (m^2/g)	Pore volume[(a)] (cm^3/g)	Amount of peptide[(b)] (mg)	Ca/P molar ratio[(c)]	ζ potential[(d)] (mV)
HAp	96	0.79	–	1.52	−24.0
p-Glu-HAp (5 mg)	107	0.91	2.4	1.52	−25.1
p-Glu-HAp (10 mg)	119	0.73	7.4	1.52	−32.7
p-Glu-HAp (20 mg)	134	0.79	14	1.55	−46.8
p-Lys-HAp (5 mg)	106	0.92	1.9	1.50	−11.2
p-Lys-HAp (10 mg)	133	0.84	5.3	1.49	+5.9
p-Lys-HAp (20 mg)	123	0.93	9.0	1.47	+19.8

[(a)] The Brunauer-Emmett-Teller (BET) surface area and pore volume were obtained using N_2 adsorption-desorption isotherms obtained by the Barrett-Joyner-Halenda (BJH) method.
[(b)] Amount of peptide in peptide-HAp was calculated by thermogravimetry (TG).
[(c)] The Ca/P molar ratios were estimated by inductively coupled plasma optical emission spectrometry (ICP-OES).
[(d)] The ζ potentials of the HAp surfaces were determined by electrophoretic light scattering method. Before the measurements, the particles were dispersed in 10 mM phosphate buffer (pH 7.0) with sonication for 3 min.

clearly show that nitrogen is widely distributed in the particle. It could therefore be considered that the presence of peptide in peptide-HAp is relatively uniform in the nanoparticles.

The presence of peptide and HAp in peptide-HAp was confirmed by FT-IR analysis as shown in Fig. 3. Fig. 3(a), for synthesized HAp, shows the stretching vibration at 3560 cm^{-1} attributed to hydroxyl groups. The bands at 1020, 960, 630, 600, and 560 cm^{-1} are attributed to the phosphate group of HAp [22–24]. For the peptide, the spectrum shows two major bands, which are amide I (−C═O stretching vibration) at 1600–1700 cm^{-1} and amide II (−N−H bending vibration and−C−N stretching vibration) at 1500–1550 cm^{-1} [25–28]. The spectrum of pure p-Glu is shown in Fig. 3(b), and the peaks at 1640 cm^{-1} correspond to amide I, while those at 1560 and 1520 cm^{-1} are attributed to amide II and that at 1400 cm^{-1} corresponds to peptide side chains. In the p-Glu-HAp (10 mg) spectrum, as shown in Fig. 3(c), bands characteristic of the hydroxyl groups and phosphate group are observed at 3560, 1020, 960, 620, 600, and 560 cm^{-1}, and the peaks at 1650, 1560, and 1400 cm^{-1} were assigned to amide I, amide II, and the side chain of p-Glu, respectively. Evidence is thus provided for the presence of HAp and p-Glu in the materials [25]. As shown in Fig. 3(d), the peak of pure p-Lys at 1610 cm^{-1} is attributed to amide I, that at 1520 cm^{-1} corresponds to amide II, and those at 1470 and 1460 cm^{-1} are attributed to the side chain of p-Lys. In the case of p-Lys-HAp (10 mg), as shown in Fig. 3(e), the peaks at 3560, 1020, 960, 620, 600, and 560 cm^{-1} correspond to the hydroxyl and phosphate groups of HAp,

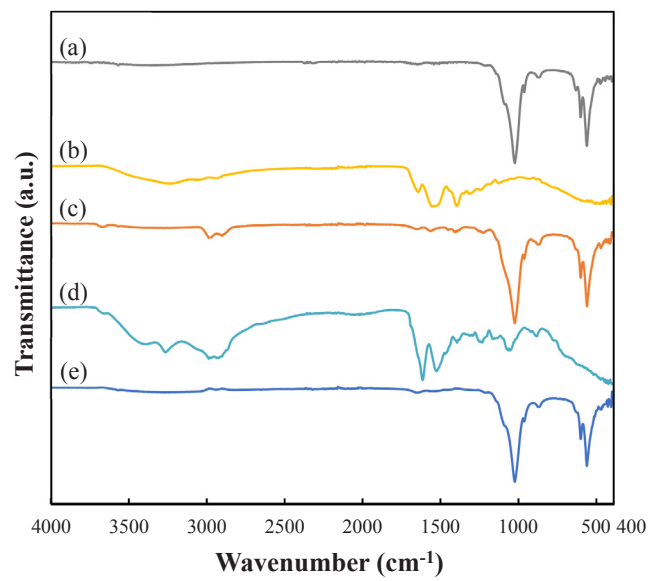

Fig. 3. FT-IR spectra of the as-synthesized HAp and peptides. (a) HAp, (b) p-Glu, (c) p-Glu-HAp (10 mg), (d) p-Lys, and (e) p-Lys-HAp (10 mg) are denoted by gray, yellow, orange, light blue, and blue, respectively.

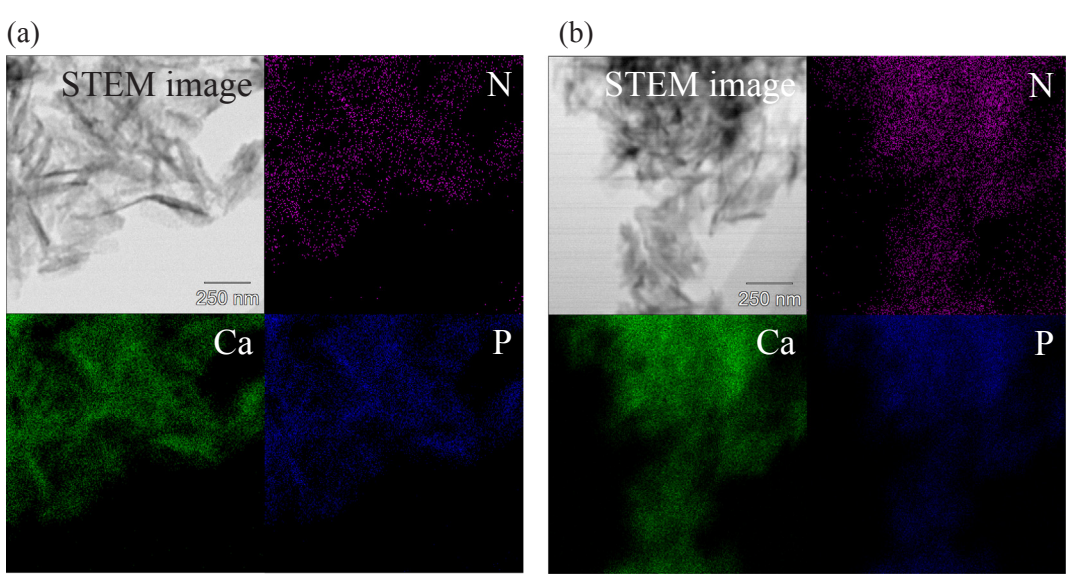

Fig. 2. STEM-EDX elemental mapping images of (a) p-Glu-HAp (20 mg) and (b) p-Lys-HAp (20 mg). Violet, green, and blue indicate nitrogen, calcium, and phosphorus, respectively.

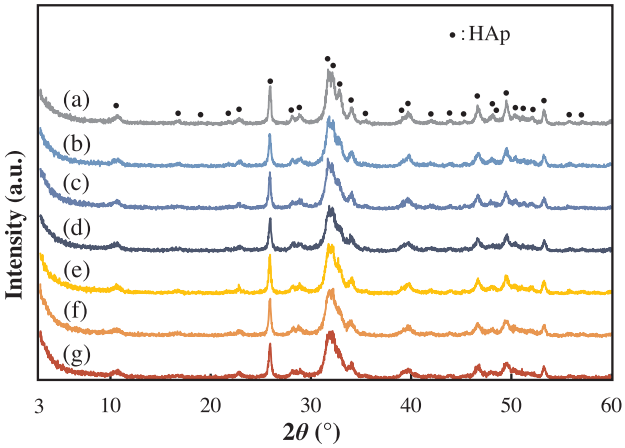

Fig. 4. XRD patterns of (a) HAp, (b) p-Glu-HAp (5 mg), (c) p-Glu-HAp (10 mg), (d) p-Glu-HAp (20 mg), (e) p-Lys-HAp (5 mg), (f) p-Lys-HAp (10 mg), and (g) p-Lys-HAp (20 mg).

and those at 1650, 1540, and 1460 cm^{-1} are attributed to amide I, amide II, and the side chain of p-Lys, respectively. These results indicate that p-Glu-HAp and p-Lys-HAp samples contain both HAp and peptide.

To estimate the amount of peptide in peptide-HAp, TG-DTA was conducted. This indicated that the weight loss in the temperature range 200–700 °C could be attributed to the decomposition of the peptide. From the exothermic peak, we calculated the amount of peptides (Table 1). The amounts of p-Glu in p-Glu-HAp (5, 10, and 20 mg) are 2.4, 7.4, and 14 mg, respectively, while in p-Lys-HAp (5, 10, and 20 mg), the amounts of p-Lys are 1.9, 5.3, and 9.0 mg, respectively. According to these data, we confirmed that the wt% of peptides in peptide-HAp was approximately 1–7.

Fig. 4 displays the XRD patterns of the HAp and peptide-HAp. For HAp, as shown in Fig. 4(a), the diffraction peaks at $2\theta = 26.0°$, 31.9°, 32.8°, and 39.8° are characteristic peaks of HAp; however, no peak was observed for the (1 0 0) reflection at $2\theta = 4.7°$ corresponding to octacalcium phosphate [25,29]. A peak at $2\theta = 26.0°$ corresponding to the (0 0 2) direction, which indicates the *c*-face of hydroxyapatite, and the one at $2\theta = 32.8°$ corresponding to the (3 0 0) direction, which indicates the *a*-face of hydroxyapatite, were observed. The two peaks retained their intensities, and it could be considered that neither the *c*-face nor the *a*-face decreased significantly [30]. The spectra of all the peptide-HAp samples, including p-Glu-HAp (5, 10, and 20 mg) and p-Lys-HAp (5, 10, and 20 mg), as shown in Fig. 4(b–g), contain the peaks corresponding to HAp at $2\theta = 26.0°$, 31.9°, 32.8°, and 39.8°, which are broader than the diffraction peaks of pure HAp due to the presence of peptide. From these results, all the calcium phosphates synthesized in this study are believed to contain low-crystallinity HAp.

The Ca/P molar ratios of HAp and peptide-HAp, which were determined by ICP-OES, are shown in Table 1. The Ca/P molar ratios of synthesized HAp, p-Glu-HAp (5, 10, and 20 mg), and p-Lys-HAp (5, 10, and 20 mg) were 1.52, 1.52, 1.52, 1.55, 1.50, 1.49, and 1.47, respectively. The Ca/P molar ratio of the resulting calcium phosphate was independent of the amount of peptides. The Ca/P molar ratios of all the synthesized calcium phosphates were lower than the stoichiometric value of HAp of 1.67; however, the XRD data indicated that the calcium phosphates synthesized in this study were low-crystallinity HAp [31,32].

To confirm the effect of peptide in particles, ζ-potential analysis was performed for all HAp particles. The surface charges of HAp and p-Glu-HAp (5 mg) were −24.0 and −25.1 mV, respectively. Furthermore, as the amount of p-Glu increased (from 10 to 20 mg), the ζ-potential changed from −32.7 to −46.8 mV. However, the surface potentials of p-Lys-HAp with different amounts of the basic peptide p-Lys (5, 10, and 20 mg) were −11.2, +5.9, and + 19.8 mV, respectively. From these results, it could be observed that the surface potential of nanoparticles was easily changed by mixing them with peptides (Table 1).

3.3. Adsorption of protein on the synthesized HAp

Cyt c, MGB, and BSA were adsorbed on HAp and peptide-HAp to investigate their protein adsorption capacities, and the results are shown in Fig. 5. The amounts of Cyt c, MGB, and BSA adsorbed on pure HAp were 60.7, 45.8, and 73.6 μg/mg, respectively. Furthermore, p-Glu-HAp containing different amounts of p-Glu (5, 10, and 20 mg) had higher Cyt c adsorption capacities of 90.4, 95.2, and 98.5 μg/mg, respectively. In contrast, the MGB and BSA adsorption capacities of p-Glu-HAp (5, 10, and 20 mg) samples decreased as the amount of peptide increased. These results revealed that the adsorption selectivity of peptide-HAp for proteins significantly improved as the amount of peptides in peptide-HAp increased.

On the contrary, for the p-Lys-HAp particles with 5, 10, and 20 mg of p-Lys, the amounts of BSA adsorbed were 95.0, 97.9, and 98.4 μg/mg, respectively, while the amounts of Cyt c and MGB adsorbed for p-Lys-HAp (5, 10, and 20 mg) were lower. Furthermore, p-Lys-HAp (20 mg) had no or extremely low adsorption capacity for both Cyt c and MGB, indicating that p-Lys-HAp (20 mg) exhibited highly selective adsorption, especially for an acidic protein such as BSA.

In conclusion, it could be confirmed that the increase in the amount

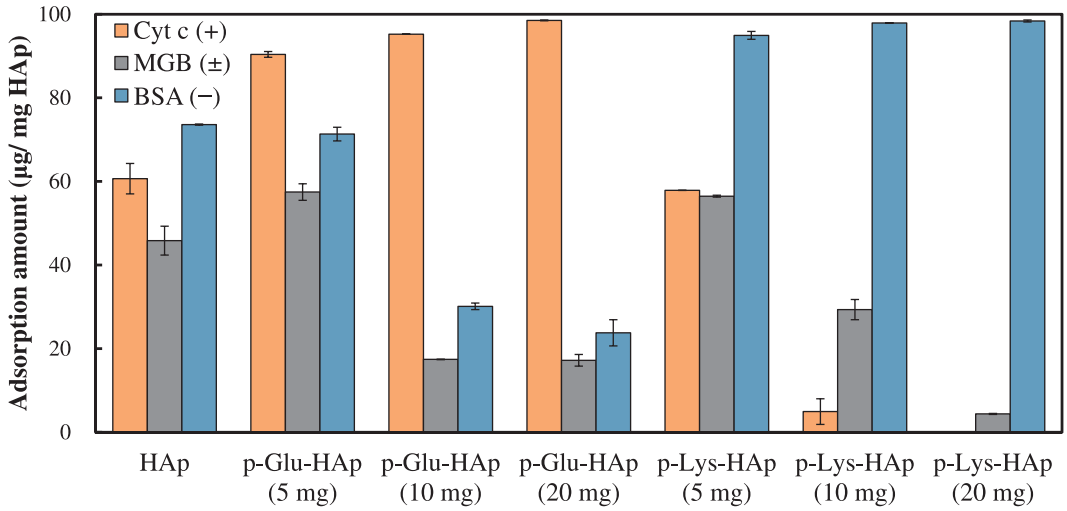

Fig. 5. Adsorbed amounts of Cyt c, MGB, and BSA on HAp, p-Glu-HAp (5, 10, and 20 mg), and p-Lys-HAp (5, 10, and 20 mg).

Table 2
Langmuir and Freundlich parameters for Cyt c on p-Glu-HAp (20 mg) and BSA on p-Lys-HAp (20 mg).

Sample	Langmuir parameters			Freundlich parameters		
	Q_m [(a)] (mg/mg)	K_L [(b)] (mL/mg)	R^2	$1/n$ [(c)]	K_F [(d)] (μg/mg)	R^2
p-Glu-HAp (20 mg)	0.22	145	0.998	0.300	0.610	0.742
p-Lys-HAp (20 mg)	0.16	156	0.999	0.249	0.0645	0.802

(a) Q_m is for the maximum amount of protein adsorbed.
(b) K_L is the equilibrium constant for the Langmuir model.
(c) $1/n$ denotes the Freundlich adsorption intensity constant.
(d) K_F is the equilibrium constant for the Freundlich model.

of peptides in peptide-HAp changed the properties of the particle surface, and the peptides in HAp induced the selective adsorption of a specific protein.

3.4. Adsorption isotherms of proteins on peptide-HAp

The Langmuir and Freundlich parameters for Cyt c on p-Glu-HAp (20 mg) and BSA on p-Lys-HAp (20 mg) are shown in Table 2. The adsorption isotherm models were estimated by fitting to Langmuir or Freundlich equations with regard to (R2) coefficients. The R^2 values for Langmuir plots for p-Glu-HAp (20 mg) and p-Lys-HAp (20 mg) were 0.998 and 0.999, which indicate fitting lines based on the Langmuir equation. The R^2 values of Freundlich plots corresponding to p-Glu-HAp (20 mg) and p-Lys-HAp (20 mg) were 0.742 and 0.802, respectively. According to these results, the R^2 values for the Langmuir plots are higher than those for the Freundlich plots; that is, each protein is adsorbed on the peptide-HAp as a monolayer. In addition, the maximum adsorption of p-Lys-HAp (20 mg) reached 160 μg/mg HAp for BSA and that of p-Glu-HAp (20 mg) for Cyt c reached 215 μg/mg HAp as shown Fig. S4.

To remove the peptides from peptide-HAp, p-Glu-HAp (20 mg) was calcined at 500 °C for 6 h. The material was added to protein solution, and adsorption was performed as described in Section 2.4. The adsorption of Cyt c on the calcined p-Glu-HAp (20 mg) decreased from 215 to 77.9 μg/mg, as shown Fig. S5. These results showed that selective adsorption significantly decreased after calcination of the p-Glu-HAp (20 mg); that is, peptide in HAp is essential for the selective adsorption of proteins.

3.5. Oxidation activity of Cyt c immobilized on HAp

To investigate the catalytic activity of adsorbed protein on HAp, the oxidation activity of Cyt c immobilized on HAp and p-Glu-HAp (10 mg) was evaluated as shown in Fig. 6. The relative activity of free Cyt c in solution was 0.121 mmol/mg/min and was considered to be 100%, while the activity of Cyt c adsorbed on HAp was 0.0780 mmol/mg/min (approximately 35% lower than that of free Cyt c). However, the activity of Cyt c adsorbed on p-Glu-HAp (10 mg) was 0.0906 mmol/mg/min, which is similar to that of Cyt c adsorbed on HAp. These results indicated that the presence of peptides in the materials does not influence the activity of Cyt c on peptide-HAp.

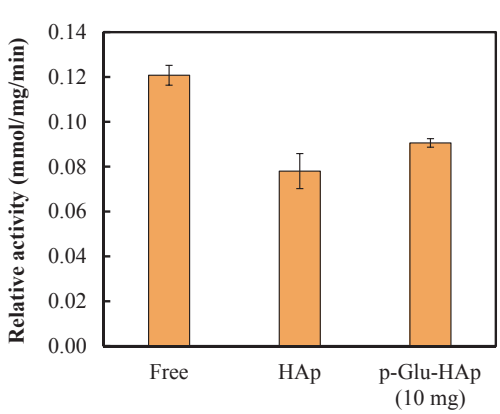

Fig. 6. Catalytic activity of Cyt c adsorbed on HAp and p-Glu-HAp (10 mg).

3.6. Conformational changes of protein adsorbed on peptide-HAp

Fluorescence spectroscopy was used to analyze the influence of the interactions between proteins and peptide-HAp, and ANS was used as a fluorescence probe. For example, when the structure of the protein adsorbed on the material was changed, the fluorescence spectrum and intensity shifted [33,34]. Fig. 7(a) shows that the fluorescence spectrum and intensity of BSA adsorbed on p-Glu-HAp (10 mg) and p-Lys-HAp (10 mg) were similar to those of free BSA. In addition, as shown in Fig. 7(b), all the peaks of Cyt c were similar. These results indicated that there were no conformational changes in the protein adsorbed on peptide-HAp.

3.7. Application of peptide-HAp particles for column chromatography and cycling

The adsorption-desorption of proteins on peptide-HAp is shown in Fig. 8(A): the supernatant of p-Glu-HAp (20 mg) after adsorption (a), p-Glu-HAp (20 mg) after desorption (b), the supernatant of p-Lys-HAp (20 mg) after adsorption (c), and p-Lys-HAp (20 mg) after desorption (d). First, for the free Cyt c and BSA mixture, each band represents a certain molecular weight (Mw). After adsorption, p-Glu-HAp (20 mg) showed a BSA band because this adsorbed only Cyt c (as shown in (a)). The adsorbed Cyt c was desorbed from p-Glu-HAp (20 mg) particles (b). With p-Lys-HAp (20 mg), the supernatant showed a band for Cyt c after adsorption (c), and the BSA was recovered after desorption (d). According to these results, synthesized peptide-HAp separated each protein from a mixed protein solution.

Cycle tests were then performed using p-Glu-HAp (20 mg) and p-Lys-HAp (20 mg), as shown in Fig. 8(B). These results showed that the capacity of protein adsorption of these materials did not change after five cycles, indicating that they are potentially suitable for protein purification.

4. Conclusion

In summary, we reported that peptide-containing HAp particles could selectively adsorb proteins. Peptide-HAp with different amounts of peptides were synthesized using simple processes. The morphology of the synthesized HAp was characterized using FE-SEM, TEM, BET surface area, and BJH pore size distribution. To confirm the presence of peptides in peptide-HAp, STEM-EDX, FT-IR, TG-DTA, and zeta-potential analysis were performed. The FT-IR, XRD, and ICP-OES results showed that the as-synthesized calcium phosphate in this study is low-crystallinity HAp. Furthermore, it was concluded that the selectivity of protein adsorption of peptide-HAp significantly improved as the amount of peptides increased. This selective adsorption of proteins was caused by electrostatic interactions between peptide-HAp and the protein, because the surface potential of peptide-HAp changed as the amount of peptides increased. Moreover, these peptide-HAp could be recycled by repeated adsorption-desorption. Therefore, peptide-HAp particles are potentially suitable as adsorbents and carrier materials for the separation of proteins.

Conflicts of interest

The authors declare no competing financial interest.

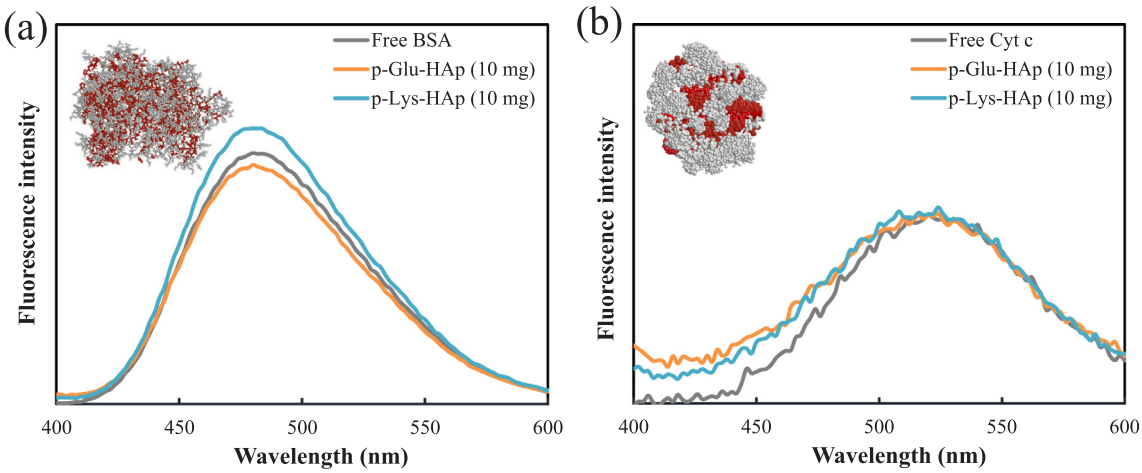

Fig. 7. Fluorescence spectrum of (a) BSA and (b) Cyt c adsorbed on p-Glu-HAp (10 mg) and p-Lys-HAp (10 mg); gray, orange, and light blue lines denote free protein, p-Glu-HAp (10 mg), and p-Lys-HAp (10 mg), respectively. Three-dimensional structures of BSA and Cyt c were obtained from the Protein Data Bank (PDB); ID: 4F5S and 2N3B, respectively. Red balls indicate the hydrophobic amino residues in the protein.

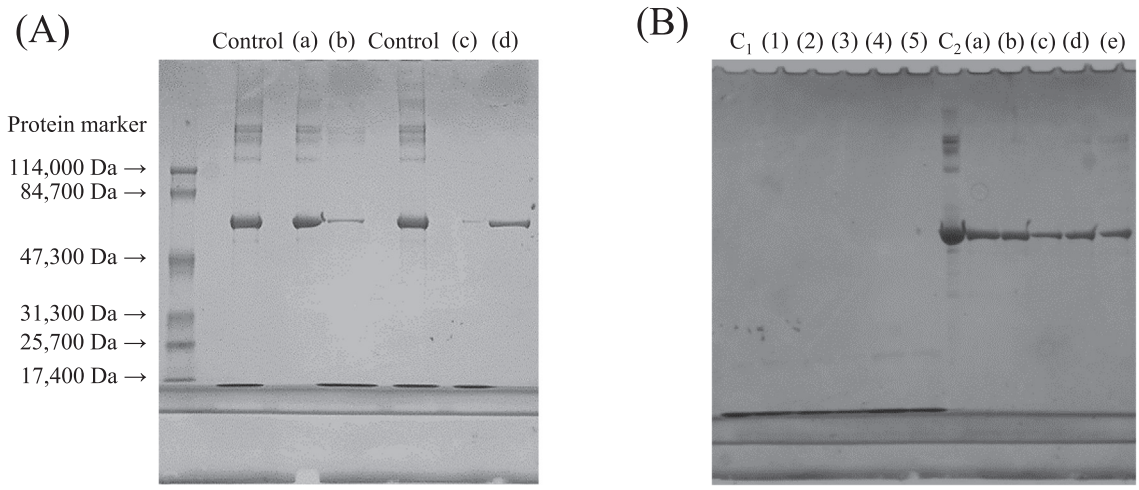

Fig. 8. (A) Adsorption-desorption of protein on peptide-HAp; the supernatant of p-Glu-HAp (20 mg) after (a) adsorption and (b) desorption, and the supernatant of p-Lys-HAp (20 mg) after (c) adsorption and (d) desorption. The control is a free Cyt c and BSA mixture (Mw of Cyt c = 12,300 Da and that of BSA = 67,000 Da). (B) Cycle testing of peptide-HAp was performed five times for adsorption-desorption; (C1) free Cyt c, (1–5) the supernatant of p-Glu-HAp (20 mg) after desorption; the numbers indicate the number of cycles of adsorption-desorption, (C2) free BSA, and (a–e) the supernatant of p-Lys-HAp (20 mg) after desorption.

Acknowledgments

We thank Dr. Hiroyuki Iwata, Associate Professor of Aichi Institute of Technology, for help with the STEM-EDX observation. This work is partly supported by Grant-in-Aid for Scientific Research (C) No. 15 K06474 from the Japan Society for the Promotion of Science (JSPS) and A-STEP (AS282I011e) from Japan Science and Technology Agency (JST).

References

[1] V.J. Mohanraj, Y. Chen, Nanoparticles: a review, Trop. J. Pharm. Res. 5 (2006) 561–573.
[2] W. Ding, N. Kameta, H. Minamikawa, M. Masuda, M. Kogiso, Preparation and formation process of Zn(II)-coordinated nanovesicles, Langmuir 33 (2017) 14130–14138.
[3] Z. Yang, J. Si, Z. Cui, J. Ye, X. Wang, Q. Wang, K. Peng, W. Chen, S. Chen, Biomimetic composite scaffolds based on surface modification of polydopamine on electrospun poly(lactic acid) / cellulose nanofibers, Carbohyd. Polym. 174 (2017) 750–759.
[4] O.V. Salata, Applications of nanoparticles in biology and medicine, J. Nanobiotechnol. 2 (2004) 3.
[5] S.A. Bhakta, E. Evans, T.E. Benavidez, C.D. Garcia, Protein adsorption onto nanomaterials for the development of biosensors and analytical devices: A review, Anal. Chim. Acta. 872 (2015) 7–25.
[6] K.S. Soppimath, T.M. Aminabhavi, A.R. Kulkarni, W.E. Rudzinski, Biodegradable polymeric nanoparticles as drug delivery devices, J. Control Release 70 (2001) 1–20.
[7] I. Lynch, K.A. Dawson, Protein-nanoparticle interactions, Nanotoday 3 (2008) 40–47.
[8] C.D. Walkey, J.B. Olsen, H. Guo, A. Emili, W.C.W. Chan, Nanoparticle size and surface chemistry determine serum protein adsorption and macrophage uptake, J. Am. Chem. Soc. 134 (2012) 2139–2147.
[9] R. Hikosaka, F. Nagata, M. Tomita, K. Kato, Effect of the pore diameters and amino-organo functional structures on mesoporous silicas for DNA adsorption, Key Eng. Mat. 720 (2017) 31–36.
[10] T. Kawasaki, S. Takahashi, K. Ikeda, Hydroxyapatite high-performance liquid chromatography: column performance for proteins, Eur. J. Biochem. 153 (1985) 361–371.
[11] M.P. Ginebra, T. Traykova, J.A. Planell, Calcium phosphate cements as bone drug delivery systems: a review, J. Control Release 113 (2006) 102–110.
[12] K. Kandori, A. Fudo, T. Ishikawa, Adsorption of myoglobin onto various synthetic hydroxyapatite particles, Phys. Chem. Chem. Phys. 2 (2000) 2015–2020.
[13] K. Kandori, K. Murata, T. Ishikawa, Microcalorimetric study of protein adsorption onto calcium hydroxyapatites, Langmuir 23 (2007) 2064–2070.
[14] Z. Zhuang, M. Aizawa, Protein adsorption on single-crystal hydroxyapatite particles with preferred orientation to *a*(*b*)- and *c*-axes, J. Mater. Sci: Mater. Med. 24 (2013) 1211–1216.
[15] C. Xu, D. He, L. Zeng, S. Luo, Colloid. Surface. B 73 (2009) 360–364.
[16] Y. Ding, J. Liu, X. Jin, H. Lu, G. Shen, R. Yu, Poly-L-lysine/hydroxyapatite/carbon nanotube hybrid nanocomposite applied for piezoelectric immunoassay of carbohydrate antigen 19–9, Analyst 133 (2008) 184–190.
[17] S. Song, S. Ren, J. Wang, S. Yang, J. Zhang, Preparation and tribological study of a

peptide-containing alkylsiloxane monolayer on silicon, Langmuir 22 (2006) 6010–6015.
[18] R. Fujisawa, Y. Wada, Y. Nodasaka, Y. Kuboki, Acidic amino acid-rich sequences as binding sites of osteonectin to hydroxyapatite crystals, Biochim. Biophys. Acta 1292 (1996) 53–60.
[19] D. Pantarotto, C.D. Partidos, J. Hoebeke, F. Brown, E. Kramer, J. Briand, S. Muller, M. Prato, A. Bianco, Immunization with peptide-functionalized carbon nanotubes enhances virus-specific neutralizing antibody responses, Chem. Biol. 10 (2003) 961–966.
[20] T. Nagasaki, F. Nagata, M. Sakurai, K. Kato, Effects of pore distribution of hydroxyapatite particles on their protein adsorption behavior, J. Asian Ceram. Soc. 5 (2017) 88–93.
[21] M.P. Ferraz, F.J. Monteiro, C.M. Manuel, Hydroxyapatite nanoparticles: a review of preparation methodologies, J. Appl. Biomater. Biom. 2 (2004) 74–80.
[22] J. Sun, L. Wu, J. Chen, Efficient lysozyme adsorption on chitosan/hydroxyapatite hybrid membrane via in situ synthesis, Cellulose 23 (2016) 3861–3874.
[23] G. Bharath, R. Madhu, S. Chen, V. Veeramani, A. Balamurugan, D. Mangalaraj, C. Viswanathan, N. Popandian, Enzymatic electrochemical glucose biosensors by mesoporous 1D hydroxyapatite-on-2D reduced graphene oxide, J. Mater. Chem. B 3 (2015) 1360–1370.
[24] C. Sarkar, P. Kumari, K. Anuvrat, S.K. Sahu, J. Chakraborty, S. Garai, Synthesis and characterization of mechanically strong carboxymethyl cellulose-gelatin-hydroxyapatite nanocomposite for load-bearing orthopedic application, J. Mater. Sci. 53 (2018) 230–246.
[25] M. Qi, J. Qi, G. Xiao, K. Zhang, C. Lu, Y. Lu, One-step hydrothermal synthesis of carbonated hydroxyapatite porous microspheres with a large and uniform size regulated by L-glutamic acid, Cryst. Eng. Comm. 18 (2016) 5876–5884.
[26] K. Kurihara, T. Abe, N. Higashi, M. Niwa, Steric forces between brush layers of poly (L-glutamic acid) and their dependence on secondary structures as determined by FT-IR spectroscopy, Colloid. Surface A 103 (1995) 265–272.
[27] Y. Yamaoki, H. Imamura, A. Fulara, S. Wójcik, Ł. Bożycki, M. Kato, T.A. Keiderling, W. Dzwolak, An FT-IR study on packing defects in mixed β-aggregates of poly(L-glutamic acid) and poly(D-glutamic acid): a high-pressure rescue from a kinetic trap, J. Phys. Chem. B 116 (2012) 5172–5178.
[28] M. Müller, Orientation of α-helical poly(L-lysine) in consecutively adsorbed polyelectrolyte multilayers on texturized silicon substrates, Biomacromolecules 2 (2001) 262–269.
[29] F. Nagata, Y. Yamauchi, M. Tomita, K. Kato, Hydrothermal synthesis of hydroxyapatite nanoparticles and their protein adsorption behavior, J. Ceram. Jpn. 121 (2013) 797–801.
[30] K. Kandori, T. Kuroda, S. Togashi, E. Katayama, Preparation of calcium hydroxyapatite nanoparticles using microreactor and their characteristics of protein adsorption, J. Phys. Chem. B 115 (2011) 653–659.
[31] L.T. de Jonge, S.C.G. Leeuwenburgh, J.G.C. Wolke, J.A. Jansen, Organic-inorganic surface modifications for titanium implant surfaces, Pharm. Res. 25 (2008) 2357–2369.
[32] S. Bose, S. Tarafder, Calcium phosphate ceramic systems in growth factor and drug delivery for bone tissue engineering: a review, Acta Biomater. 8 (2012) 1401–1421.
[33] K. Kato, K. Inukai, K. Fujikura, T. Kasuga, Effective encapsulation of laccase in an aluminum silicate nanotube hydrogel, New J. Chem. 38 (2014) 3591–3599.
[34] K. Nakanishi, M. Tomita, H. Nakamura, K. Kato, Specific binding of immunoglobulin G to protein A-mesoporous silica composites for affinity column chromatography, J. Mater. Chem. B 1 (2013) 6321–6328.